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ProcDNA Case Study: Nexavir | PrimeRx Flow

Alert-to-Treatment & Commercial Opportunity Analysis

2026-27

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PrimeRx Flow – Business & Case Study Overview

Business Overview:

- **PrimeRx** is a pharmaceutical company operating across the U.S. healthcare landscape, and its flagship product is **Nexavir** (DRG001), a treatment designed for high clinical need and broad prescriber adoption.
- You have been assigned to analyze PrimeRx's Rx flow, HCP, account, sales, and alert datasets collected over the last 18 months and provide recommendations to help the sales team act on the most valuable commercial opportunities.

Case Study Overview:

This case study explores a pharma commercial analytics problem: **doctors (HCPs) are diagnosing patients who may need treatment, but not all of them are prescribing our drug**.

You will use clinical alert, doctor affiliation, and prescription sales data to identify:

- HCPs and accounts with high prescription and alert activity
- Doctors receiving strong clinical signals but writing few or no prescriptions
- Accounts with low prescription activity relative to their doctor base
- HCPs and accounts that should be prioritized by the sales team

The goal is to connect **diagnostic activity** with **prescribing behavior** and translate the insights into a clear, data-driven action plan

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PrimeRx Flow – Datasets Overview

You are provided with three datasets capturing different aspects of the patient treatment journey:

- **Alerts**: Clinical signals indicating potential treatment need at the doctor level
- **Sales**: Prescription activity across doctors and therapies
- **Affiliation**: Mapping of doctors to hospitals/healthcare accounts

  These datasets can be linked using **HCP_ID**. Your task is to connect clinical activity with prescribing behavior to identify missed opportunities and recommend actionable strategies for the commercial team.

Dataset	Rows	Key Columns	Dataset Purpose
Alerts	9,537	Alert_ID (unique alert), Alert_Date, HCP_ID, Lab_Result (positive/negative)	Clinical alerts tied to doctors indicating potential treatment need
Sales	5,917	Prescription_ID (unique Rx event), Prescription_Date, HCP_ID, Drug_Name, Drug_ID, Prescription_Volume (units/scripts prescribed)	Prescription activity across doctors and drugs
Affiliation	216	HCP_ID, Account_ID, Account_Name	Maps doctors to hospitals/clinics/accounts

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Question 1: Who are the most active doctors and are they helping our product?

Your starting universe is only the doctors who appear in the **Alerts** dataset i.e., doctors for whom at least one alert has been generated. Do not include any doctor who has no alert record.
From this universe, produce two separate Top 10 lists using two different ranking metrics:

- **Ranking 1 - Top 10 by Alert Volume**: Which 10 doctors have the highest total number of alerts generated?
- **Ranking 2 - Top 10 by Prescription Volume:** Which 10 doctors **from the alerts universe** have written the highest prescription volume in the Sales data?

Ranking 1 - Top 10 by alert volume

Rank	HCP_ID	Total Alerts
1	HCP041	200
2	HCP011	196
3	HCP035	194
4	HCP029	187
5	HCP021	186
6	HCP044	180
7	HCP042	174
8	HCP036	171
9	HCP027	165
10	HCP010	163

Ranking 2 - Top 10 by prescription volume

Rank	HCP_ID	Total Prescription Volume
1	HCP185	2900
2	HCP019	2736
3	HCP192	2710
4	HCP187	2680
5	HCP188	2650
6	HCP183	2620
7	HCP189	2590
8	HCP186	2560
9	HCP184	2530
10	HCP190	2500

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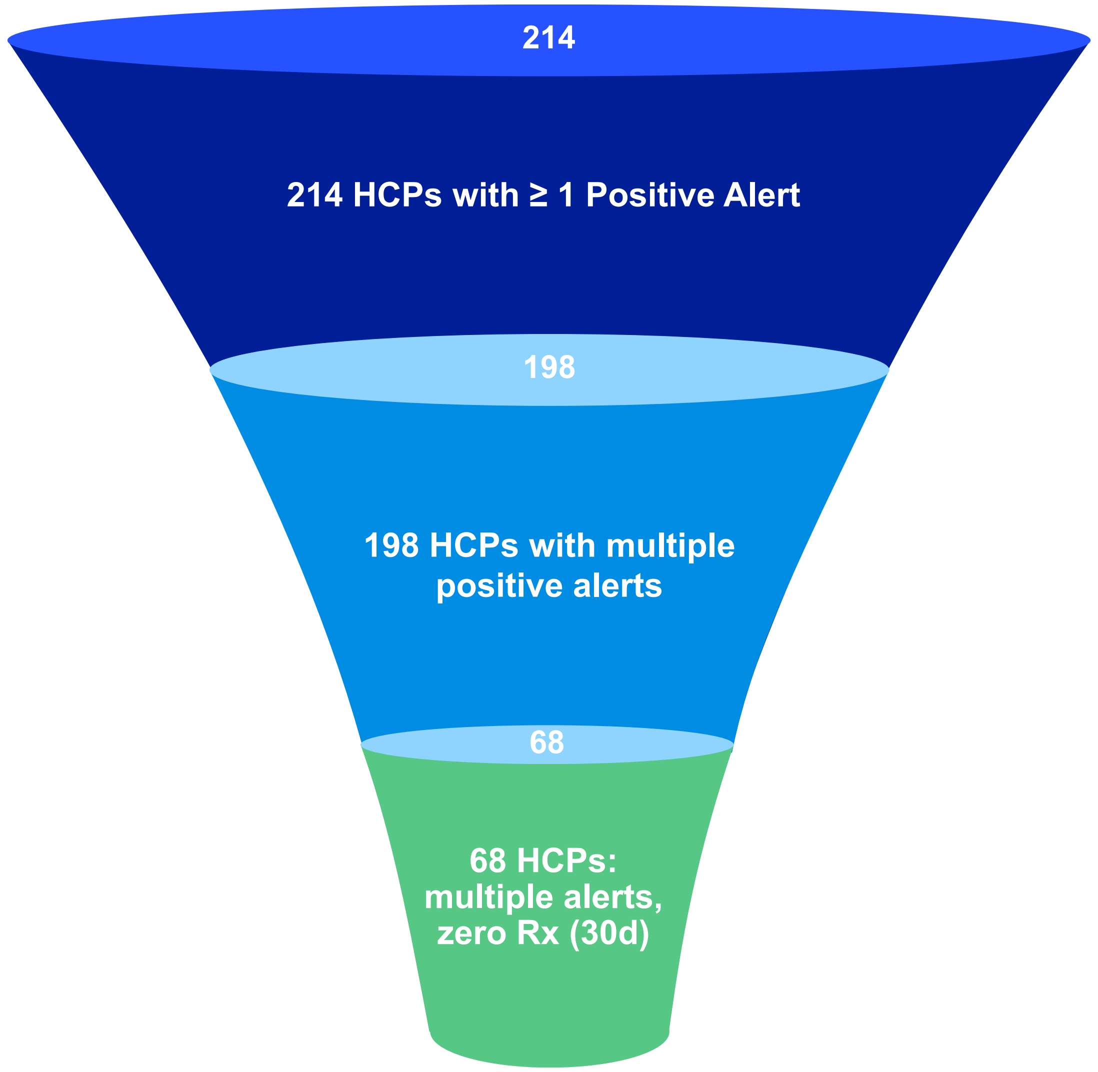
Question 2: Which doctors see the need but don't act on it?

Identify all doctors from the alerts universe who received **positive lab test alerts**, indicating that the patient likely required treatment. Evaluate whether these doctors showed any prescription activity for **any drug within 30 days** following their **first positive alert**. Among doctors with no prescribing activity during this post-alert window, rank the Top 10 HCPs by total **positive alert volume**.
Additionally, create a funnel showing HCPs with at least one positive alert, multiple positive alerts, and multiple positive alerts but zero prescribing activity within 30 days.

Positive-alert doctors and 30-day prescription follow-up

Rank	HCP_ID	Positive Alerts
1	HCP041	127
2	HCP011	118
3	HCP035	112
4	HCP029	108
5	HCP021	104
6	HCP044	99
7	HCP042	96
8	HCP036	93
9	HCP027	89
10	HCP010	85



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Question 3: Among our top doctors, how much business is going to competitors?

Use the **Top 10 doctors by total alert volume** from Question 1.
For these 10 doctors, calculate the share of total prescription volume contributed by:

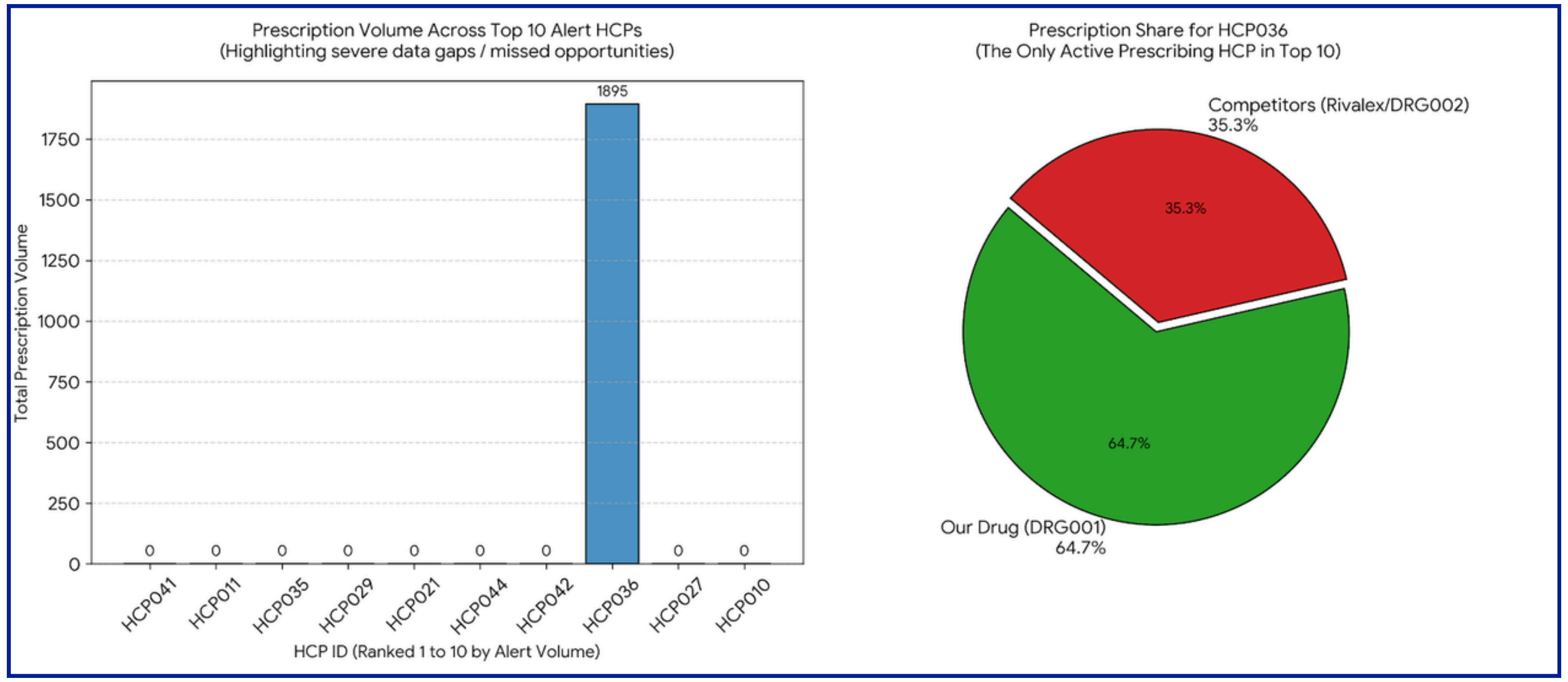
- **Our Drug (DRG001)**
- **Competitor Drugs (all other drugs)**
  Next, identify which competitor drug accounts for the highest prescription volume and is capturing the largest share among your highest-alerting doctors.

Prescription mix for Top 10 alert-volume HCPs

Alert Rank	HCP_ID	Total Prescription Volume	% Our Drug Share	% Competitor Share
1	HCP041	0	0.0%	0.0%
2	HCP011	0	0.0%	0.0%
3	HCP035	0	0.0%	0.0%
4	HCP029	0	0.0%	0.0%
5	HCP021	0	0.0%	0.0%
6	HCP044	0	0.0%	0.0%
7	HCP042	0	0.0%	0.0%
8	HCP036	1895	64.7%	35.3%
9	HCP027	0	0.0%	0.0%
10	HCP010	0	0.0%	0.0%

Most prescribed competitor drug among these HCPs

Competitor Drug Name	Drug_ID	Total Prescription Volume
Rivalex	DRG002	1470



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Question 4: Which hospitals are sitting on the biggest untapped opportunity?

Using the Affiliation dataset, identify the **Top 10 accounts with the highest alert volume** based on activity generated by their affiliated doctors.
For these Top 10 accounts, calculate:

- Total Alert Count
- Our Drug Prescription Count (DRG001)
- Conversion Ratio - defined as the proportion of prescription count relative to alert count
  Rank the accounts by lowest conversion ratio to identify high-alert accounts with weak prescription conversion.

Top 10 accounts by alert count

Rank	Account_ID	Total Alerts	Our Drug Prescription Count	Conversion Ratio
1	ACC001	3296	12297	3.73
2	ACC003	1195	4947	4.14
3	ACC002	2350	12223	5.20
4	ACC004	530	5402	10.19
5	ACC005	585	9148	15.64
6	ACC006	493	9887	20.05
7	ACC012	85	1804	21.22
8	ACC009	100	2699	26.99
9	ACC007	114	7072	62.04
10	ACC010	92	9616	104.52

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Question 5: Which accounts represent the largest untapped prescription opportunity?

Using the Affiliation dataset, determine the **total number of doctors** associated with each account. Using prescription activity for Our Drug, calculate:

- The number of **active prescribers** within each account
- Total **prescription volume** at the account level

  Finally, rank accounts by the **lowest prescriptions-per-active-doctor ratio** and identify the Top 10 accounts.

  Additionally, create a scatter plot comparing account size vs prescriptions-per-active-doctor ratio, highlighting potential underperforming accounts.

Account size vs company-drug sales efficiency

Rank	Account_ID	# Affiliated Doctors	# Active Doctors	Our Drug Prescription Volume	Prescription Volume per Active Doctor
1	ACC017	3	1	59	59.00
2	ACC015	8	7	1347	192.43
3	ACC012	10	7	1804	257.71
4	ACC009	11	10	2699	269.90
5	ACC003	24	15	4947	329.80
6	ACC018	4	3	1095	365.00
7	ACC004	17	13	5402	415.54
8	ACC008	7	6	3049	508.17
9	ACC014	6	6	3126	521.00
10	ACC023	1	1	573	573.00

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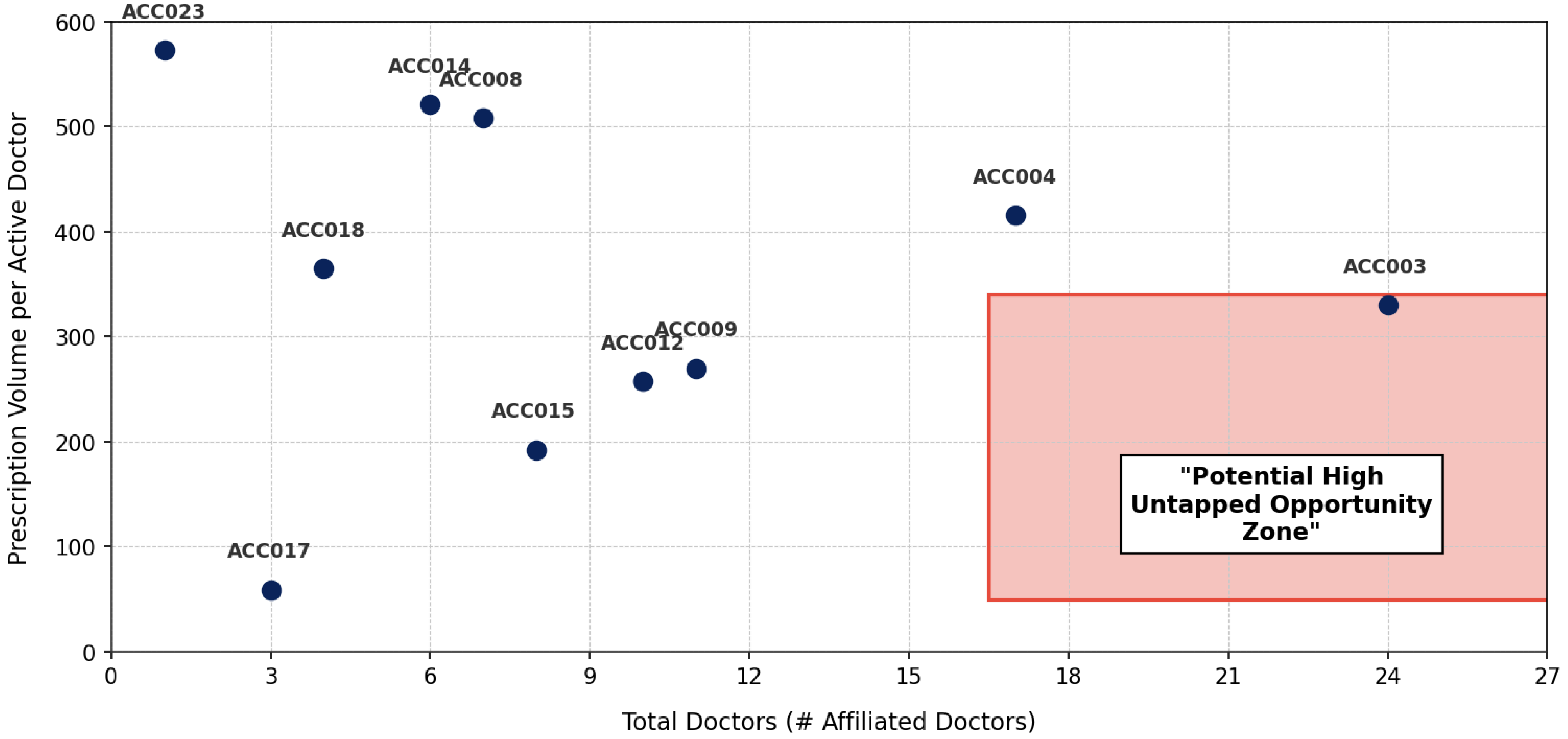
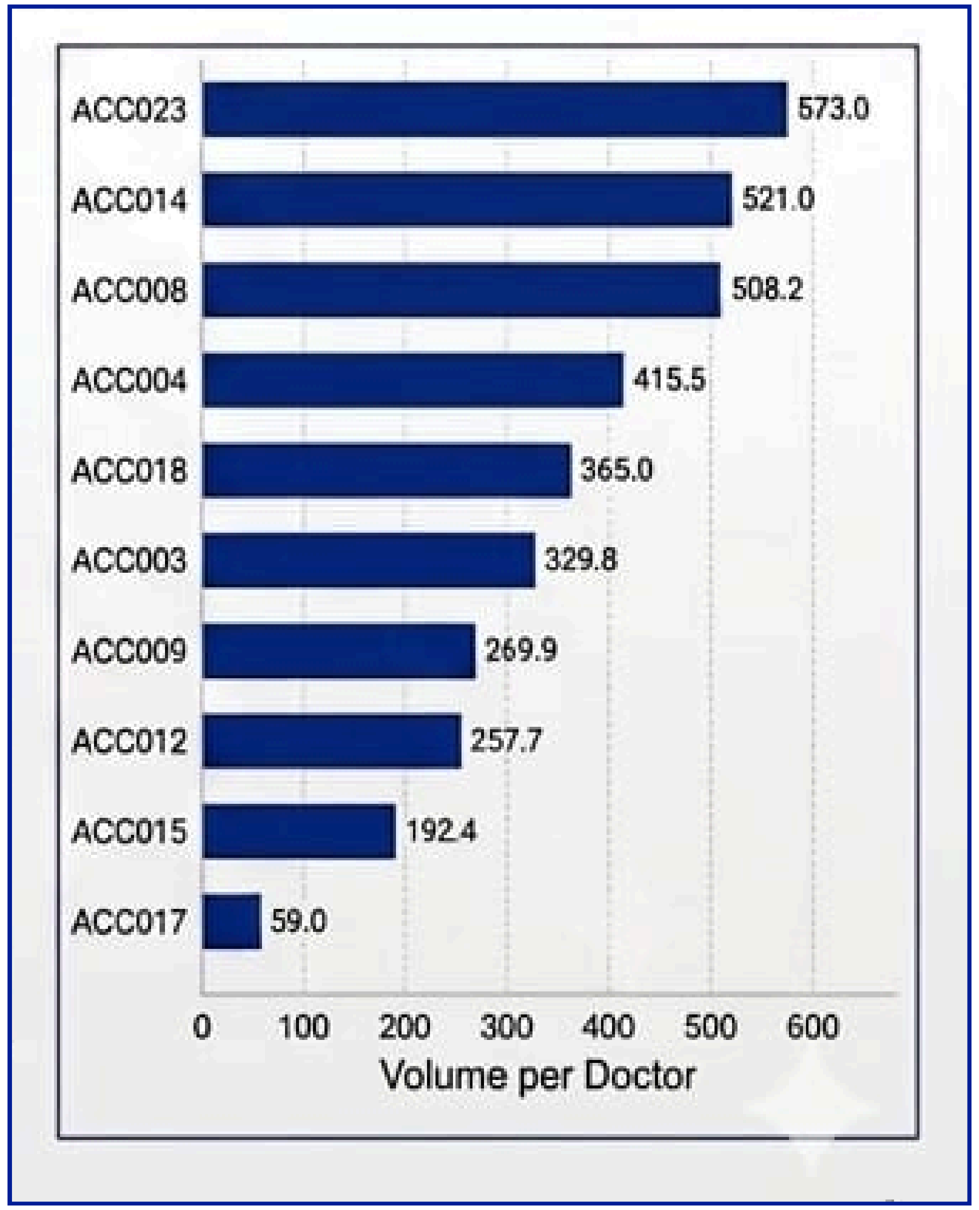
PRESCRIPTION VOLUME PER ACTIVE DOCTOR vs. TOTAL DOCTORS (# AFFILIATED DOCTORS) SCATTER PLOT





Strategic Inferences

1. ACC001 (Northside General) has the highest alert volume but lowest conversion (3.73) - top priority for intervention.
2. ACC017 (Maplewood Clinic) has only 1 active prescriber out of 3 doctors - activation opportunity.
3. Large accounts (ACC003, ACC004) show low Rx/doctor ratios despite many affiliated doctors.
4. Smaller accounts like ACC010, ACC007 show very high conversion - may serve as best-practice models.
5. Focus field force on high-alert, low-conversion accounts for maximum ROI.

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Question 6: Did alerts actually change doctor behavior - before vs. after?

Part A - Basic lift: Define a 90-day before and 90-day after window around each doctor's first positive alert. Calculate prescription volume for our drug in each window. Rank Top 10 HCPs by lift % with highest lift being Rank 1?
Part B - Segment the lift: Split all doctors into 3 behavioral bucket as mentioned in the formula table. Does the lift look different across the three groups? Which group shows strongest commercial impact?

Part A: HCP-level before / after lift

Rank	HCP_ID	Total Prescription Volume Before Alert	Total Prescription Volume After Alert	Lift %
1	HCP040	8	143	1687.5%
2	HCP050	10	107	970.0%
3	HCP006	32	249	678.1%
4	HCP108	31	238	667.7%
5	HCP016	50	279	458.0%
6	HCP159	42	232	452.4%
7	HCP157	72	387	437.5%
8	HCP211	42	223	431.0%
9	HCP209	64	335	423.4%

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Part B: Segment Summary

Segment	HCPs	Avg Lift
New starters	64	N/A (pre=0)
Growers	25	+322.8%
Non-responders	125	-25.1%
Total	**214**	

Segment showing Strongest Commercial Impact	New Starters (64 HCPs: 0 to 9,417 total Rx units)

Formula

Lift% : (Post Prescription Volume – Pre Prescription Volume) / (Pre Prescription Volume) * 100

- New starters: pre=0, post>0
- Growers: pre>0 and lift% ≥ +20%
- Non-responders: Everything else

Question 7: Commercial HCP Prioritization

A pharma company assign sales representatives to promote the drug to HCP. Your company's commercial team is planning the **next quarter's** outreach strategy for **DRG001.** Following are the field force assumptions:

- Each sales rep can complete **~8 effective HCP calls per day**
- Each rep works **~20 field days per month**
- The organization has **40 sales reps nationally**

  Leadership wants to use this limited **field force capacity** efficiently by prioritizing the **HCP segments** identified in **Question 6** (New Starters, Growers, and Non-Responders).

  Using the segment-level insights from Question 6, recommend how the company should allocate total rep call capacity across these segments.

  In your response:

- Estimate how much **outreach focus** each segment should receive
- Explain the reasoning behind your **allocation approach**
- Discuss the tradeoff between **growth opportunity, conversion likelihood, and sales effort** required across segments

Total rep call capacity per quarter
40 reps x 8 calls/day x 20 days/month x 3 months = 19,200 quarterly calls

Segment	HCPs	Suggested % of Rep Calls	Estimated Quarterly Calls	Estimated Rep Allocation
New starters	64	50%	9,600	~20
Growers	25	35%	6,720	~14
Non-responders	125	15%	2,880	~6
Total	**214**	**100%**	**19,200**	**40**

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Executive Summary

PrimeRx Nexavir (DRG001) - Commercial Analysis Summary

- **KEY FINDINGS:**
- 1. ALERT-TO-RX GAP: 68 high-alert HCPs received multiple positive clinical signals but show zero prescription activity within 30 days - the largest immediate commercial opportunity.
- 2. COMPETITOR THREAT: Rivalex (DRG002) is the top competitor capturing significant share among our most clinically active doctors.
- 3. ACCOUNT OPPORTUNITY: ACC001 (Northside General) generates 3,296 alerts but has only 3.73 conversion ratio - the single biggest untapped hospital account.
- 4. ALERT IMPACT IS REAL: 64 New Starters went from 0 to active prescribing post-alert. HCP040 showed 1,687.5% lift - proving alerts drive behavior change.
- 5. FIELD FORCE: Recommend 50% of 19,200 quarterly rep calls to New Starters, 35% to Growers, 15% to Non-responders for maximum ROI.

- **STRATEGIC RECOMMENDATIONS:**
- - Deploy targeted rep visits to 68 zero-Rx positive-alert HCPs immediately
- - Launch competitive detailing program focused on Rivalex accounts
- - Prioritize ACC001 and lowest-conversion accounts for account-level interventions
- - Scale alert-to-rep follow-up program - data proves it changes prescribing behavior

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Thank You!

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